



A Smart-Skin-Based System for Real-Time Injury Risk Detection on a Biomechanical Mannequin

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Abstract: Injury prevention in military and industrial environments requires reliable systems capable of detecting biomechanical stress under practical monitoring conditions. Conventional assessment approaches are often limited to post-event analysis and lack the ability to provide immediate feedback under dynamic loading conditions. This study aimed to develop an embedded-oriented smart-skin system for injury-risk detection using a biomechanical mannequin equipped with multisensory technology. The proposed system consisted of a flexible smart-skin layer embedded with pressure and temperature sensors and connected to an embedded data acquisition unit for continuous monitoring. Sensor data were processed through a structured pipeline comprising signal filtering, window-based segmentation, and statistical feature extraction. Several lightweight machine-learning models, including Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and eXtreme Gradient Boosting (XGBoost), were employed to classify the biomechanical conditions as either safe or injury risk. This classification supported rapid decision-making under controlled experimental conditions. Experimental evaluation was conducted under controlled static, dynamic, and combined loading scenarios. The results showed that the proposed system achieved high classification performance, with a maximum accuracy of 99.66% and consistently high F1-scores across all evaluated models. The high performance can be attributed to the use of discriminative statistical features and the controlled experimental setup, which enabled clear separation between the two classes. These findings indicate that the integration of multi-sensor smart-skin technology with efficient data processing and lightweight machine-learning models provides a feasible framework for injury-risk detection under controlled mannequin-based experimental conditions. This study contributes an integrated experimental measurement and computational analysis pipeline for embedded-oriented injury-risk monitoring, with potential applications in ergonomic assessment, military training, and occupational safety.

Keywords: Smart skin; Injury risk detection; Biomechanical mannequin; Real-time monitoring; Embedded system; Pressure sensors; Temperature sensors; Machine learning

1 Introduction

The increasing demand for safety and performance optimization in military and industrial environments has driven the development of intelligent sensing systems capable of monitoring human-machine interactions in real time. In high-risk operational settings, such as military training and heavy equipment operation, prolonged mechanical loading and uneven pressure distribution can lead to fatigue, discomfort, and an increased risk of injury [1]. However, conventional assessment methods are typically limited to post-event analysis or subjective evaluation, making them less effective for early risk detection and prevention [2].

Recent advances in smart sensing technologies, particularly electronic skin (e-skin), have enabled the development of systems that mimic the sensing capabilities of human skin by capturing physical stimuli such as pressure, force, and temperature [3, 4]. These systems offer significant potential for continuous monitoring of biomechanical stress and have been widely explored in applications such as robotics, healthcare, and human-machine interfaces [5–7]. In

addition, material innovations, including flexible polymers and hydrogel-based sensors, have improved sensitivity, flexibility, and durability, enabling more reliable sensing performance in practical environments [8, 9].

From a computational perspective, sensor-based systems often rely on machine-learning techniques to interpret complex and nonlinear signal patterns. Previous studies have demonstrated the effectiveness of methods such as Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and gradient boosting algorithms in handling high-dimensional sensor data and improving classification performance [10–12]. However, many existing studies focus primarily on algorithmic performance under controlled laboratory conditions and give limited consideration to system-level integration, real-time processing capabilities, and deployment feasibility.

Despite the rapid progress in e-skin technology and machine learning, several challenges remain. First, most studies emphasize material characterization or isolated sensing performance rather than the development of an integrated end-to-end system. Second, the translation of sensor measurements into actionable risk indicators remains limited, particularly in applications related to injury prevention. Third, real-time implementation on computationally constrained embedded platforms remains insufficiently explored. These limitations highlight the need for a practical and deployable system that integrates sensing, processing, and decision-making within a unified framework.

To address these challenges, this study proposes a smart-skin-based system for real-time injury-risk detection using a biomechanical mannequin. The system integrated pressure and temperature sensors within a flexible smart-skin layer and employed an embedded data-acquisition unit for continuous monitoring. Sensor signals were processed through a structured pipeline involving filtering, segmentation, and statistical feature extraction. Lightweight machine learning models were used to classify biomechanical conditions as either safe and injury risk, thereby enabling rapid and interpretable decision-making.

The main hypothesis of this study was that an integrated multisensor smart-skin system combined with efficient signal processing and lightweight machine-learning models can reliably distinguish between safe and injury-risk conditions in real time under controlled loading scenarios. Experimental validation was conducted using static, dynamic, and combined loading conditions on a biomechanical mannequin representing a typical operator profile.

The main contribution of this work is the development and validation of a practical, real-time smart-skin system that bridges sensor-based measurement and computational analysis. Unlike studies that focus solely on algorithmic improvements, this research emphasizes system-level feasibility, real-time operation, and the interpretability of results. The findings showed that the proposed approach could provide a reliable foundation for injury-risk monitoring in applications such as military training, ergonomic assessment, and occupational safety.

2 Methodology

2.1 The Proposed System

The proposed system was designed to perform real-time injury-risk detection using a smart-skin-based sensing platform integrated with a biomechanical mannequin. As illustrated in Figure 1, the system consists of three main components: (1) a multisensor smart-skin layer, (2) an embedded data-acquisition unit, and (3) a computational processing pipeline for classification.

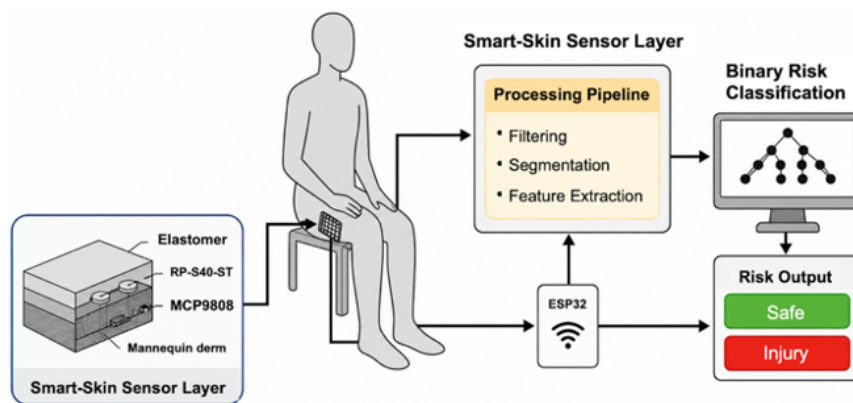


Figure 1. Architecture of the proposed smart-skin-based system for real-time injury-risk detection

The smart-skin layer incorporated pressure sensors (RP-S40-ST) and digital temperature sensors (MCP9808), embedded within a flexible elastomer material to mimic the mechanical characteristics of human skin behavior. Sensor signals were acquired using an ESP32-based microcontroller and were subsequently transmitted for further processing. The overall system workflow, as shown in Figure 1, comprised signal acquisition, preprocessing (filtering and segmentation), statistical feature extraction, and classification as either safe or injury risk.

2.2 Data Acquisition and Experimental Setup

The dataset used in this study was obtained from controlled experiments conducted on a biomechanical mannequin representing a typical human operator profile. The mannequin was used as a stable experimental platform to ensure repeatable sensor placement and consistent loading conditions during data acquisition. Pressure and temperature data were recorded under three types of loading scenarios: static loading, dynamic loading, and combined loading.

As shown in the Table 1, the smart-skin prototype consisted of RP-S40-ST pressure sensors and MCP9808 digital temperature sensors mounted on the surface of the mannequin. The sensors were attached using adhesive fixation and covered with a silicone elastomer layer to improve surface conformity and mechanical stability during loading. The silicone coating thickness was approximately 0.14 mm, while the total thickness measured from the mannequin surface to the outermost silicone-coated layer was approximately 90 mm. The sensing nodes were positioned in the lateral torso region of the mannequin to capture localized pressure and temperature variations under controlled loading conditions. Figure 2 shows the physical placement of the smart-skin sensors on the mannequin.

Table 1. Physical configuration of the smart-skin prototype

Parameter	Description
Pressure sensor	RP-S40-ST
Temperature sensor	MCP9808
Coating material	Silicone elastomer
Silicone coating thickness	0.14 mm
Total thickness from mannequin surface to outer skin layer	90 mm
Mounting method	Adhesive fixation on mannequin surface
Data acquisition unit	ESP32-based acquisition unit
Sensor placement	Lateral torso region of the mannequin
Measured variables	Pressure and temperature
Window size	100 samples
Window overlap	50%



Figure 2. Physical placement of the smart-skin sensor nodes on the biomechanical mannequin

Meanwhile, the experimental loading protocol can be seen in Table 2.

Table 2. Experimental loading protocol

Loading Condition	Mechanical Load Condition	Temperature Condition	Time	Repetition	Expected Response
Static Loading	Constant pressure applied to the sensing area	Stable ambient temperature	30 s	5 trials	Stable pressure and temperature response
Dynamic Loading	Periodically applied fluctuating pressure	Stable ambient temperature	30 s	5 trials	Temporal pressure variation with moderate thermal response
Combined Loading	Simultaneous sustained and fluctuating pressure	Increased localized thermal exposure	30 s	5 trials	Elevated pressure and temperature responses indicative of an injury-risk condition

The experimental loading protocol was designed to simulate biomechanical stress conditions associated with prolonged contact and repeated mechanical interaction. Static loading was used to evaluate the sensor response under sustained pressure conditions, while dynamic loading represented repeated or fluctuating mechanical stress. Combined loading was introduced to simulate simultaneous mechanical and thermal stress conditions associated with an elevated injury risk. Each experimental condition was repeated five times to ensure measurement consistency and repeatability.

In addition, the injury-risk labeling scheme is summarized in Table 3. The labeling process was based on calibrated threshold conditions derived from the observed relationship between pressure response and localized temperature variation during the controlled loading experiments.

Table 3. Injury-risk label definition

Class	Pressure Response	Temperature Response	Description
Safe	Low to moderate pressure variation	Stable temperature response	Conditions within acceptable biomechanical and thermal limits
Injury	Elevated or sustained pressure response	Increased localized temperature response	Conditions exceeding predefined injury-risk thresholds

All borderline samples near the decision threshold were assigned according to the dominant pressure and temperature responses within the corresponding window to maintain consistent labeling across the dataset.

Sensor data were acquired using an ESP32-based data-acquisition unit and recorded continuously during each experimental session. To capture temporal variations in biomechanical stress, the recorded signals were segmented into fixed-length windows of 100 samples with a 50% overlap. Each window represented a short-duration snapshot of the biomechanical state and was used as an input sample for feature extraction and classification.

2.3 Signal Preprocessing and Feature Extraction

Raw sensor signals are often affected by noise and fluctuations caused by mechanical and environmental disturbances; therefore, preprocessing was applied to improve signal quality prior to feature extraction. This preprocessing stage included basic filtering to remove high-frequency noise and normalization to ensure consistent scaling across different the sensor channels.

Each segmented window was subsequently transformed into a set of statistical features representing its temporal characteristics. The extracted features included the mean, standard deviation, root mean square (RMS), skewness, kurtosis, peak value, and peak-to-peak amplitude. These features captured both amplitude-based and distribution-based properties of the signals, providing an effective representation of biomechanical stress patterns while reducing the data dimensionality required for subsequent processing.

The mean value of each window was calculated as:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i \quad (1)$$

where, x_i represents the signal value of the i -th sample, and N is the number of samples per window. The standard deviation was calculated as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (2)$$

Meanwhile, the RMS value was calculated as:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (3)$$

These statistical features provided a compact yet informative representation of the underlying signal behavior, thereby supporting efficient and reliable classification in the subsequent stage.

2.4 Classification Models

To perform injury-risk detection, several lightweight machine-learning models were employed, including Random Forest, SVM with a radial basis function (RBF) kernel, KNN, and eXtreme Gradient Boosting (XGBoost). These models were selected because of their computational efficiency, robustness, and suitability for real-time implementation on resource-constrained embedded systems.

Random Forest was used as the primary model because of its robustness to noise and its capability to model nonlinear relationships within high-dimensional feature spaces [13–15]. SVM was included because of its effectiveness in constructing nonlinear decision boundaries [16], while KNN was employed as a distance-based classification approach suitable for low-complexity pattern recognition [17]. XGBoost was employed to evaluate the effectiveness of a boosting-based ensemble-learning method with regularization capability and efficient optimization [18]. The selected models were configured using lightweight parameter settings to maintain a balance between classification accuracy and computational efficiency. The configurations of the evaluated models are summarized in Table 4.

Table 4. Machine learning model configuration

Model	Configuration
Random Forest	100 decision trees using Gini impurity criterion
SVM (RBF)	Radial basis function kernel with default regularization parameters
KNN	Euclidean distance metric with $k = 5$
XGBoost	100 estimators with learning rate of 0.1

Note: SVM = Support Vector Machine; RBF = radial basis function; KNN = K-Nearest Neighbors; XGBoost = eXtreme Gradient Boosting.

The selected model configurations were designed to maintain low computational complexity while preserving reliable classification performance, supporting the feasibility of real-time implementation on embedded sensing platforms.

2.5 Model Training and Evaluation

The dataset was divided into training and testing sets using a holdout strategy, with 70% of the data used for training and 30% used for testing. Stratified splitting was applied to preserve the class distribution of the safe and injury-risk classes in both subsets, thereby maintaining consistent class proportions during model evaluation. Model training and testing were performed under controlled experimental conditions using the extracted statistical features as model inputs.

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Accuracy is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

where, TP , TN , FP , and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. Precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

The F1-score, which represents the harmonic mean of precision and recall, is defined as:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

In addition to the numerical performance metrics, a confusion matrix was used to analyze the classification behavior and identify potential misclassification patterns between the safe and injury-risk classes. This analysis provided insight into the ability of the proposed system to reliably distinguish between safe and elevated biomechanical stress conditions. The evaluation framework was designed to assess both classification performance and computational feasibility for embedded-oriented implementation. Because the experiments were conducted under controlled loading conditions, the obtained results primarily reflected the capability of the proposed smart-skin system to differentiate between biomechanical states within the experimental setup.

2.6 Implementation Systems

The proposed system was designed as an embedded-oriented injury-risk detection framework. The use of statistical feature extraction and lightweight machine-learning models reduces computational complexity, thereby making the processing pipeline potentially suitable for implementation on resource-constrained platforms such as ESP32-based acquisition systems.

In this study, the ESP32 was used primarily as the data-acquisition unit, while model training and evaluation were conducted offline using the extracted feature dataset. Therefore, the reported classification results should be interpreted as evidence of computational feasibility rather than as the results of a full embedded deployment benchmark. Further work is required to evaluate inference latency, memory usage, power consumption, and real-time execution performance on the target embedded hardware.

All data used in this study were obtained from controlled experimental measurements. The dataset is available from the corresponding author upon reasonable request, subject to institutional policies.

3 Results

3.1 Dataset Characteristics

The dataset used in this study comprised of 980 samples obtained from controlled experimental measurements using a biomechanical mannequin integrated with a smart-skin sensing system. Each sample represented a segmented window of sensor data containing 100 samples and a 50% overlap, thereby capturing short-term biomechanical conditions.

The dataset was categorized into two classes: safe and injury. As shown in Table 5, the dataset was relatively balanced, with 517 samples labeled as safe and 463 labeled as injury. This balanced distribution helped limit class imbalance and supported the evaluation of model performance.

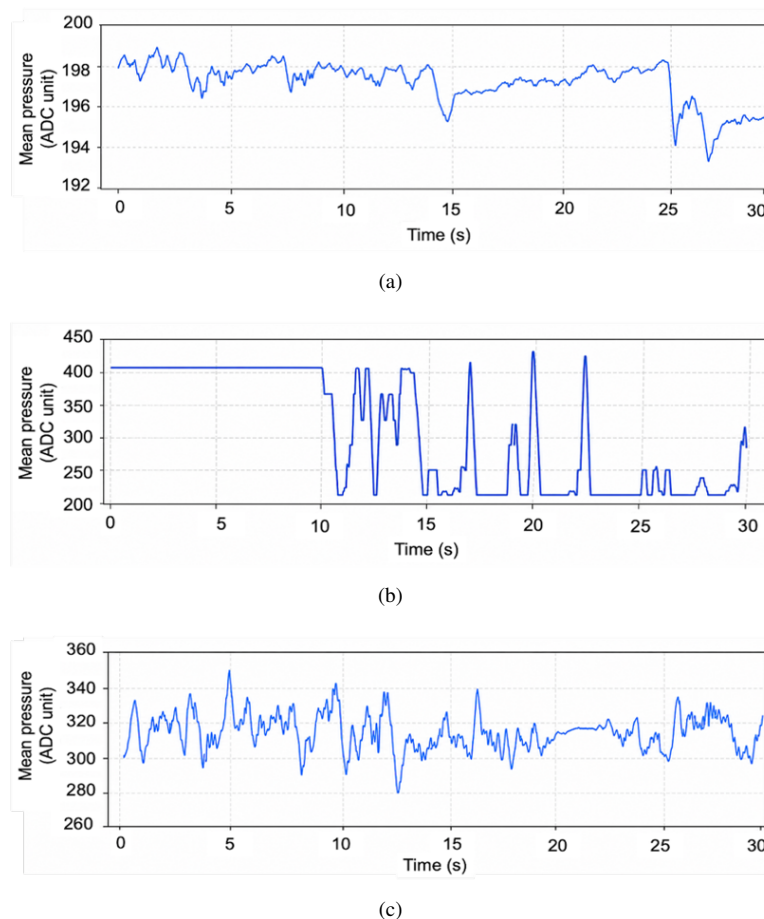
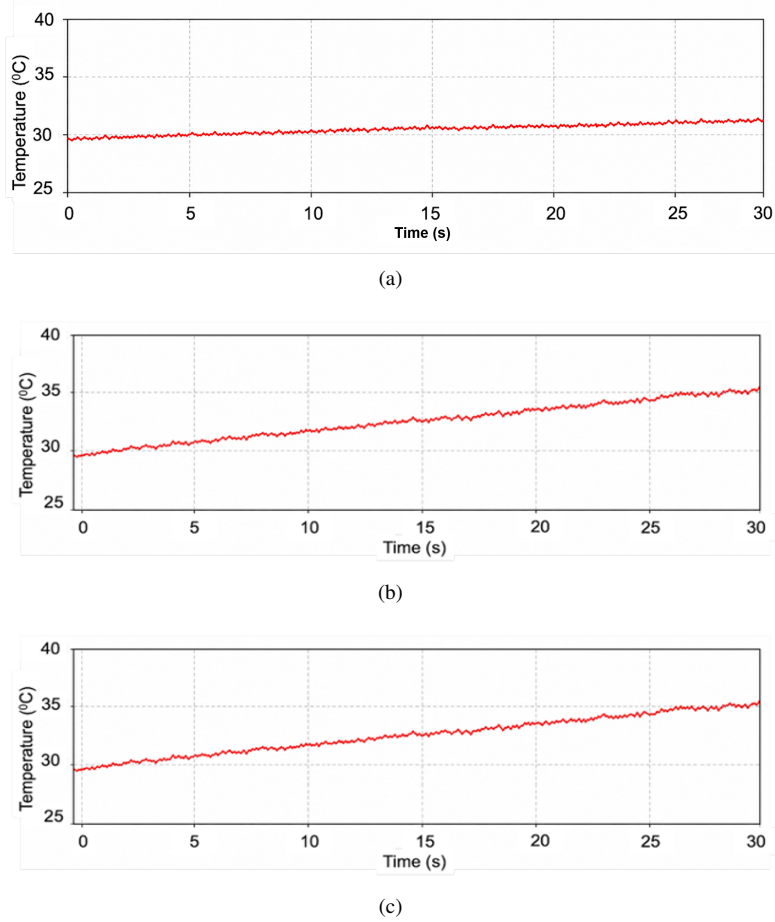


Figure 3. Representative pressure time-series responses under: (a) static loading condition; (b) dynamic loading condition; (c) combined loading condition

Note: ADC = Analog-to-Digital Converter; the dynamic loading condition shows abrupt fluctuations and plateau-like responses associated with repeated mechanical contact, while the combined loading condition exhibits higher variability due to simultaneous sustained and fluctuating pressure.

Table 5. Class distribution of the dataset

Class	Number of Samples
Safe	517
Injury	463
Total	980

**Figure 4.** Representative temperature time-series responses under: (a) static loading condition; (b) dynamic loading condition; (c) combined loading condition

Note: The temperature response corresponds to the recorded temperature channel.

Figure 3 presents representative pressure time-series responses obtained under the three experimental loading conditions: static loading, dynamic loading, and combined loading. The pressure response was calculated as the mean value of sensors A–E. The static loading condition exhibited relatively stable pressure behavior, whereas the dynamic loading condition showed abrupt fluctuations and plateau-like responses associated with repeated mechanical contact. The combined loading condition exhibited greater variability resulting from simultaneous sustained and fluctuating pressure.

Figure 4 presents representative temperature time-series responses under the same loading conditions. The temperature response was obtained from the recorded temperature channel. The static loading condition exhibited a relatively stable thermal response with a gradual increase, while the dynamic and combined loading conditions showed more pronounced temperature increases. This trend suggested that combined mechanical and thermal exposure produced a higher thermal response than static loading.

The window-based segmentation approach was applied to transform continuous sensor recordings into structured samples for feature extraction and classification. Because adjacent windows overlapped by 50%, temporally neighboring samples could share some signal information. Therefore, the reported results should be interpreted in the context of the controlled experimental evaluation, and their generalization to independent sessions or real-world conditions requires further validation.

3.2 Feature Analysis

Statistical features extracted from each segmented window were used to represent the temporal characteristics of the pressure and temperature signals. These features include mean, standard deviation, RMS, skewness, kurtosis, peak value, and peak-to-peak amplitude. The combination of these features enabled the representation of both amplitude-based and distribution-based properties of the signals.

Amplitude-related features such as mean, RMS, and peak values generally exhibit higher magnitudes for the injury class compared to the safe class. This behavior reflected increased mechanical loading and pressure intensity under injury-risk conditions. As illustrated in Figure 5, the distributions of mean and RMS features for the injury class tended to shift toward higher values compared to the safe class. In contrast, the safe class generally showed lower median values, although several outliers were observed, indicating that some safe windows still contained localized signal fluctuations.

Distribution-based features, including skewness and kurtosis, provided additional information regarding the shape and variability of the signal distribution. These features helped capture nonlinear characteristics and transient fluctuations that were not fully represented by amplitude-based metrics alone. As a result, they contributed to improved separability between the safe and injury classes.

Overall, the extracted statistical features provided a compact and informative representation of biomechanical stress patterns. The observed differences between the two classes, supported by the feature distributions shown in Figure 5, indicated that the selected features are suitable for distinguishing between safe and injury-risk conditions, supporting the effectiveness of the subsequent classification process.

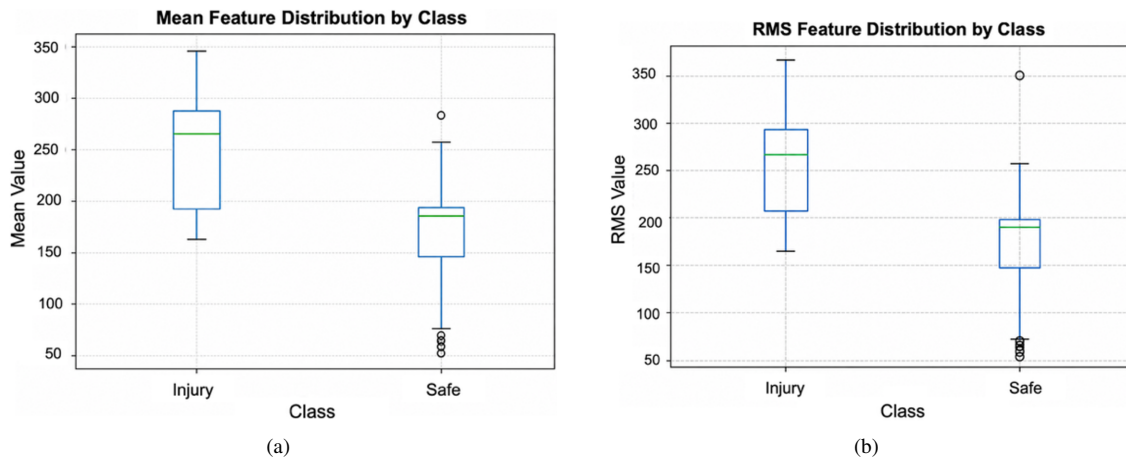


Figure 5. Distribution of statistical features between safe and injury conditions: (a) mean feature; (b) root mean square (RMS) feature

3.3 Model Performance Evaluation

The performance of the classification models was evaluated using accuracy, precision, recall, and F1-score. Four machine-learning models were tested, including Random Forest, SVM, KNN, and XGBoost. The performance results are summarized in Table 6.

Table 6. Model performance comparison for injury risk detection

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.9932	0.9858	1.0000	0.9929
SVM (RBF)	0.9864	0.9720	1.0000	0.9858
KNN ($k = 5$)	0.9898	0.9857	0.9928	0.9892
XGBoost	0.9966	1.0000	0.9928	0.9964

Note: SVM = Support Vector Machine; RBF = radial basis function; KNN = K-Nearest Neighbors; XGBoost = eXtreme Gradient Boosting.

All evaluated models demonstrated high classification performance, with accuracy values exceeding 98%. Among the evaluated models, XGBoost achieved the highest overall performance, followed closely by Random Forest. These results indicated that ensemble-based learning methods effectively captured nonlinear relationships within the extracted statistical features.

To further analyze the classification results, the confusion matrix of the best-performing model, XGBoost, is presented in Figure 6. The model correctly classified 155 safe samples and 138 injury samples. One injury sample was misclassified as safe, whereas no safe samples were misclassified as injury. This result was consistent with the high precision and recall values reported in Table 6.

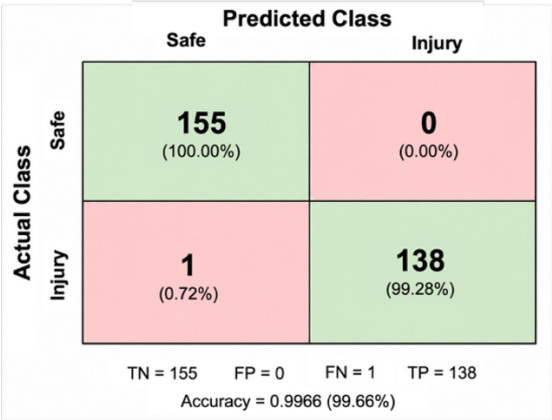


Figure 6. Confusion matrix of the eXtreme Gradient Boosting (XGBoost) model
 Note: TN = true negative; FP = false positive; FN = false negative; TP = true positive.

The high classification performance can be associated with the controlled loading protocol and the discriminative capability of the extracted statistical features, particularly amplitude-related features such as the mean and RMS values. However, because the dataset was obtained from controlled mannequin-based experiments using overlapping-window segmentation, the reported results should be interpreted in the context of the experimental setup. Further validation under independent sessions and real-world conditions is required to evaluate the broader generalization of the system.

To further evaluate the classification capability of the proposed system, a receiver operating characteristic (ROC) analysis was performed for the best-performing model. As shown in Figure 7, the XGBoost classifier achieved an area under the curve (AUC) value of 0.9987, indicating excellent separability between the safe and injury classes. The ROC analysis confirmed that the extracted statistical features provided strong discriminative capability for injury-risk detection within controlled experimental setup.

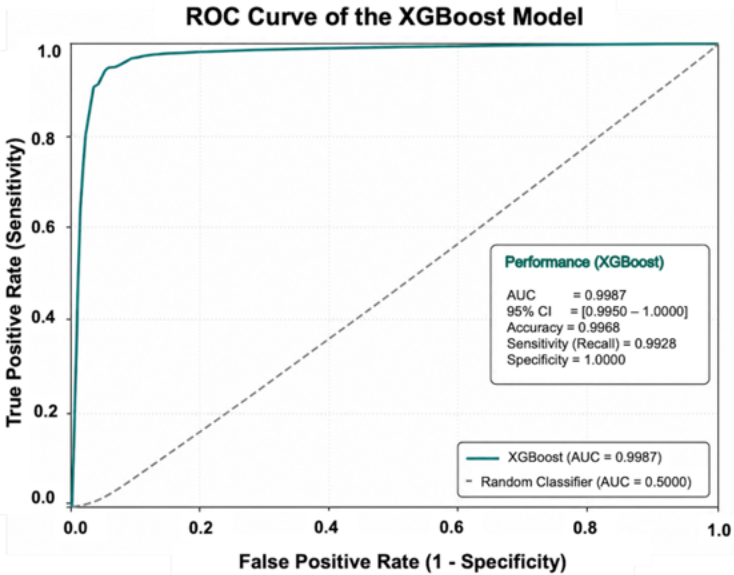


Figure 7. Receiver operating characteristic (ROC) curve of the eXtreme Gradient Boosting (XGBoost) model for injury-risk classification
 Note: AUC = area under the curve.

Overall, the experimental results showed that the proposed smart-skin-based system could distinguish between safe and injury-risk conditions using computationally efficient machine-learning models. The combination of

multisensor data acquisition, statistical feature extraction, and lightweight classification methods provided a reliable framework for embedded-oriented injury-risk monitoring under controlled experimental conditions.

4 Discussion

4.1 Interpretation of Classification Performance

The experimental results showed that all evaluated models achieved high classification performance, with accuracy values exceeding 98%. Among the models, XGBoost and Random Forest exhibited the most consistent performance across all evaluation metrics. This high performance may be attributed to the use of discriminative statistical features, particularly amplitude-based features such as the mean and RMS, which effectively captured variations in biomechanical loading conditions.

In addition, the controlled experimental setup contributed to the clear separation between the safe and injury-risk classes. Because the data were collected under well-defined loading scenarios, the resulting feature distributions exhibited distinct patterns, thereby facilitating accurate classification. However, the high performance should be interpreted in the context of a controlled mannequin-based experiment, in which the loading conditions and label boundaries were experimentally defined. Therefore, the results primarily indicated the feasibility of the proposed sensing and classification pipeline rather than its broad generalization to uncontrolled real-world environments.

4.2 Analysis of Misclassification

Despite the high overall performance, a small number of misclassifications were observed, as indicated by the confusion matrix. These misclassifications are primarily associated with borderline conditions where the distinction between safe and injury states is less pronounced.

Such cases may have resulted from transitional biomechanical conditions or slight variations in sensor responses under similar loading levels. Additionally, the use of overlapping windows during segmentation may have introduced similarities between adjacent samples, which could have contributed to minor classification errors. This condition may also have partially explained the near-perfect performance because neighboring windows could share similar temporal information. Nevertheless, the low number of misclassifications indicated that the system maintained a high level of reliability in distinguishing between the two classes within the controlled experimental setting.

4.3 System-Level Implications

The results highlighted the effectiveness of the proposed smart-skin-based system for injury-risk detection in a controlled experimental setting. The integration of multisensor data acquisition, statistical feature extraction, and lightweight machine-learning models enables efficient processing with low computational overhead.

The use of models such as Random Forest and XGBoost demonstrated that high performance can be achieved without requiring complex deep-learning architectures. This suggests that the system is potentially suitable for implementation on embedded platforms, such as ESP32-based systems, where computational resources are limited. However, because the machine learning models were evaluated offline in this study, the actual embedded inference latency, memory usage, and power consumption must still need to be measured before full real-time deployment can be claimed. Consequently, the proposed approach has strong potential for practical implementation in applications such as ergonomic monitoring, military training, and occupational safety, provided that further validation is performed under realistic operational conditions.

4.4 Comparison with Previous Studies

Compared with previous studies that focus primarily on material development or sensor-level analysis, this work emphasizes system-level integration and real-time applicability. Many existing studies on e-skin systems concentrate on sensor design, material properties, or isolated sensing performance without addressing full-system deployment or real-time processing challenges [8, 9, 19].

From a computational perspective, several studies have explored machine-learning and deep-learning approaches for analyzing smart-skin or sensor data, often relying on complex models or large-scale datasets [20]. In contrast, the proposed system showed that effective injury-risk detection could be achieved using compact feature representations and lightweight classification models, thereby making the approach more suitable for embedded-oriented implementation.

The obtained results were comparable to or exceeded those reported in related studies, particularly under controlled environments. However, direct comparison remains limited due to differences in experimental setups, sensor configurations, and evaluation protocols across studies. Therefore, the main contribution of this work should be regarded as an integrated experimental measurement and computational pipeline rather than a direct performance benchmark against previous studies.

4.5 Limitations

Several limitations should be acknowledged in this study. First, the dataset was obtained from controlled experiments using a biomechanical mannequin, which may not fully represent the variability among human subjects in real-world conditions. Second, the evaluation was performed using a holdout validation strategy, which may have produced to optimistic performance estimates, particularly when overlapping-window segmentation is applied.

Furthermore, environmental variations and external disturbances were limited in the experimental setup, which may affect the generalization of the system when deployed in more complex real-world scenarios. Third, the safe and injury labels were defined based on experimentally calibrated threshold conditions, and therefore the classification results reflected the ability of the models to learn the resulting controlled label structure. Fourth, embedded execution was not fully benchmarked in terms of latency, memory usage, and power consumption. In addition, practical sensing factors such as the consistency of smart-skin contact, sensor displacement during prolonged use, and the response time of the temperature sensors were not thoroughly assessed in this study. These factors could influence the measurement of pressure and temperature under more realistic operating conditions.

4.6 Future Work

Future work will focus on extending the proposed system to real-world applications by incorporating data from human subjects and a wider range of operating conditions. Additional sensor modalities, such as vibration and motion sensors, may be integrated to enhance the robustness of the system.

Furthermore, advanced validation strategies, such as cross-session and cross-subject evaluation, can be explored to better assess the generalization of the system. The integration of adaptive or learning-based threshold mechanisms may also improve the system's ability to handle dynamic and unpredictable environments. Future studies should evaluate the embedded inference performance directly on the target hardware to quantify processing time, memory consumption, and energy requirements for real-time deployment.

5 Conclusions

This study presented a smart-skin-based system for real-time injury-risk detection using a biomechanical mannequin equipped with pressure and temperature sensors. The proposed system combines multisensor data acquisition, statistical feature extraction, and lightweight machine learning models within a unified processing pipeline.

Experimental results showed that the system could accurately distinguish between safe and injury-risk conditions, with all evaluated models achieving strong classification performance. Among the tested models, XGBoost and Random Forest exhibited the most consistent results across multiple evaluation metrics. The findings showed that the selected statistical features effectively captured biomechanical stress patterns, thereby enabling reliable classification using computationally efficient methods.

The proposed approach emphasized system-level feasibility rather than algorithmic complexity, showing that high-performance injury-risk detection could be achieved without relying on resource-intensive models. The results demonstrate the feasibility of embedded-oriented injury-risk monitoring using lightweight machine-learning models under controlled experimental conditions.

Despite the promising results, this study was limited by the use of controlled experimental data obtained from a biomechanical mannequin. Future work will focus on validating the system using data from human subjects and a wider range of real-world conditions to improve its generalization performance and robustness.

Author Contributions

Conceptualization, G.A.M.; methodology, G.A.M., M.R.A., and K.P.S.S.; software, K.P.S.S.; validation, G.A.M., K.P.S.S., and S.P.R.; formal analysis, K.P.S.S.; investigation, K.P.S.S., S.P.R., and M.R.A.; resources, G.A.M. and M.R.A.; data curation, K.P.S.S., M.R.A., and S.P.R.; writing—original draft preparation, G.A.M., S.P.R., and K.P.S.S.; writing—review and editing, G.A.M.; visualization, K.P.S.S.; supervision, G.A.M.; project administration, G.A.M. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that generative artificial intelligence (AI) tools were used to assist in language refinement and drafting of certain parts of the manuscript. All scientific content, including the study design, data analysis, results, and conclusions, was developed and verified by the authors. The authors take full responsibility for the accuracy, originality, and integrity of the work. No AI tools were used to generate or manipulate data, results, or references.

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Nomenclature

x_i	Sensor signal value at the i -th sample
N	Number of samples in each window
RMS	Root mean square value
TP	True positive
TN	True negative
FP	False positive
FN	False negative

Greek symbols

μ	Mean value (statistical)
σ	Standard deviation

Subscripts

i	Index of sample
t	Time index